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Section - 4 Group - 1

Newton's Ring

Date of experiment: **Partner's name:**

Date of Lab. report submission:

Aim of the Experiment

To study the interference pattern and determine the radius of curvature of the planoconvex lens using Newton's rings apparatus.

Apparatus Required

(i) Source of light (Hg lamp)

(ii) Newton's ring apparatus

(iii) Power supply for Hg CS/50 W lamp

(iv) Interference filters (Green & yellow)

(v) Mounted lens & Screen

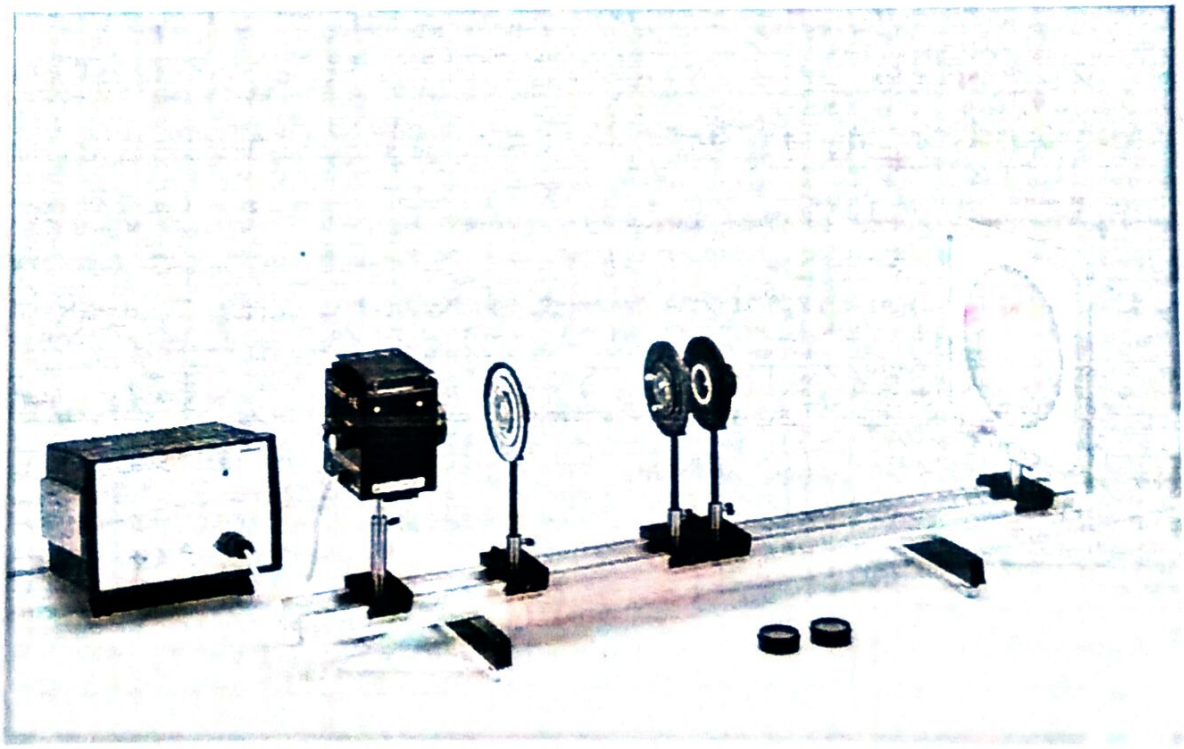


Figure-1: Experimental set up for Newton's ring.

Theory:

In Newton's rings experiment, a parallel beam of monochromatic light is incident normally on a combination of a planoconvex lens L and a glass plate G as shown in Fig.1. A part of each incident ray is reflected from the lower surface of the lens. The other part after refracting through the air film between the lens and the glass plate is reflected back from the glass plate. These two reflected rays from coherent sources of light. As a result, they interfere and produce a system of alternate dark and bright rings with the point of contact between the lens and the plate as the centre. These rings are known as Newton's rings. Looking at the diagram we can write,

$r_m^2 = R^2 - (R - t)^2$ or $2t = r_m^2 / R$ When $2Rt >> t^2$ or $2R >> t$

Where r_m is the radius of the mth bright fringe,

R is the radius of the curvature of the planoconvex lens,

t is the thickness of the air film .

For bright fringes $2t\mu \cos\theta = m\lambda$

Where μ= refractive index of the medium

θ= refraction angle

λ=wavelength of the monochromatic source

m= order of the fringe.

As we know, μ=1 for air and θ to be very small, we have

$$2t = m\lambda \quad \text{or} \quad r_m^2 / R = m\lambda$$

From this relationship one can obtain the wavelength of light if the radius of curvature of the plano-convex lens is known.

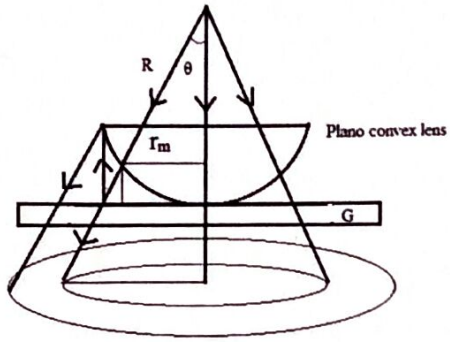
From the above expression we can write $D_m^2 = 4r_m^2 = 4Rm\lambda$.

If D_n is the diameter of the $(n)^{th}$ bright ring then $D^2_n = 4Rn\lambda$.

So we can write,

$$R = \frac{D_n^2 - D_m^2}{4(n-m)\lambda} \text{ for } n>m$$

From this the value of "***R***" the radius of curvature of the planoconvex lens can be determined.



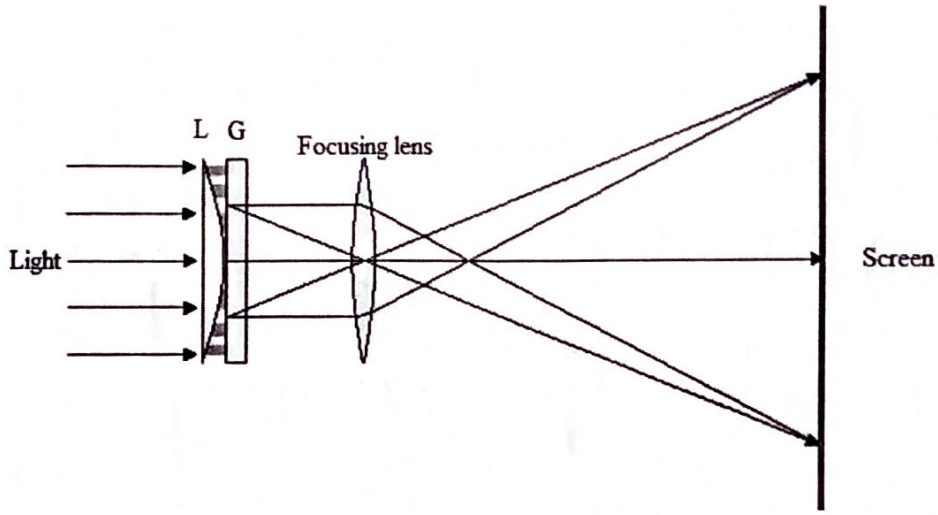


Figure-2: Schematic diagram of Newton's ring.

Procedure:

(i) All the requisite apparatuses including the mercury spectral lamp is mounted on the optical bench.

(ii) Switch on the universal choke of the power supply and wait for the warm-up for a few minutes.

(iii) Adjust the position of lens and Newton's ring apparatus to obtain coloured rings on the screen.

(iv) Clamp the green filter in the object holder.

(v) Measure the diameter of mth and n^{th} ring on the screen.

(vi) Repeat the measurement with blue colour filter.

(vii)Plot a graph between radii of the ring as a function of order number for various wavelengths.

Observation:

Table-1

Sl. No.	Colour of the filter & its wavelength	Diameter of mth ring (D_m)	Diameter of nth ring (D_n)	$D_n^2 - D_m^2$	R (m)
1	**YELLOW (578 nm)**	$D_2 = 1.04$	$D_5 = 1.44$	0.992	14.302
2		$D_4 = 1.32$	$D_7 = 1.60$	0.8176	11.787
3		$D_6 = 1.52$	$D_9 = 1.84$	1.0752	15.501

Table-2

Sl. No.	Colour of the filter & its wavelength	Diameter of mth ring (D_m)	Diameter of nth ring (D_n)	$D_n^2 - D_m^2$	R (m)
1	**GREEN (546 nm)**	$D_1 = 0.9$	$D_4 = 1.26$	0.7776	11.868
2		$D_3 = 1.16$	$D_6 = 1.46$	0.786	11.996
3		$D_5 = 1.36$	$D_8 = 1.66$	0.906	13.828

Table-3

Sl. No.	Radius of ring	Green	Yellow
1	r_1	2.7/6 = 0.45	2/5 = 0.4
2	r_2	3.2/6 = 0.53	2.6/5 = 0.52
3	r_3	3.5/6 = 0.58	3/5 = 0.60
4	r_4	3.8/6 = 0.63	3.3/5 = 0.66
5	r_5	4.1/6 = 0.68	3.6/5 = 0.72
6	r_6	4.4/6 = 0.73	3.8/5 = 0.76
7	r_7	4.7/6 = 0.78	4/5 = 0.8
8	r_8	5.0/6 = 0.83	4.3/5 = 0.86
9	r_9	5.2/6 = 0.87	4.6/5 = 0.92
10	r_{10}	5.4/6 = 0.9	4.8/5 = 0.96

Calculation:

Wave length of the filter = _____ nm

For example, we will select 25th and 20th rings from the pattern then the formula for calculating radius of curvature will be

$$R = \frac{D_{25}^2 \quad D_{20}^2}{4\,(25-20)\,\lambda}$$

Conclusion:

Conclusion should incorporate

(ix) Interesting results you found in the experiment.

(x) Lacunas /difficulties that you encountered.

(xi) Practical applications/implications of the experiment.

(xii) Any other issues.

Precautions

(i) One should carefully handle the Newton's ring set up while moving the mounted screws.

References

1. Fundamentals of optics by F. Jenkins and H. White.
2. Optics by A. Ghatak.
3. Introduction to Optics by E. Hecht
4. Feynman Lectures.

Considerations

(i) What is interference of light and what are the different ways in which light interferes?

(ii) How interference takes place in case of a Newton's ring?

(iii) Why the fringe pattern obtained from Newton's ring is circular and can it be straight?

Calculation

For yellow filter,

1 mm in original scale = 5 mm in magnified scale

Hence 5 is the dividing factor for yellow filter.

For green filter,

1 mm in original scale = 6 mm in magnified scale.

Hence 6 is the dividing factor for green filter.

For yellow filter, $\lambda = 578\ nm$

$$R_1 = \frac{D_5^2 - D_2^2}{4(5-2)\lambda} = \frac{0.992 \times 10^{-4}}{4 \times 3 \times 578 \times 10^{-9}} = 14.302\ m$$

$$R_2 = \frac{D_7^2 - D_4^2}{4(7-4)\lambda} = \frac{0.8176 \times 10^{-4}}{4 \times 3 \times 578 \times 10^{-9}} = 11.787\ m$$

$$R_3 = \frac{D_9^2 - D_6^2}{4(9-6)\lambda} = \frac{1.0752 \times 10^{-4}}{4 \times 3 \times 578 \times 10^{-9}} = 15.501\ m$$

$$\overline{R} = \frac{R_1 + R_2 + R_3}{3} = \frac{(14.302 + 11.787 + 15.501)}{3}\ m$$

$$= 13.863\ m$$

$$d_1 = \overline{R} - R_1 = (13.863 - 14.302)\ m = -0.439\ m$$

$$d_2 = \overline{R} - R_2 = (13.863 - 11.787)\ m = 2.076\ m$$

$$d_3 = \overline{R} - R_3 = (13.863 - 15.501)\ m = -1.638\ m$$

$$s^2 = \frac{\Sigma d_i^2}{n} = \frac{(-0.439)^2 + (2.076)^2 + (-1.638)^2}{3}$$

$$= 2.3952$$

$$\sigma = \sqrt{\frac{s^2}{n-1}} = \sqrt{\frac{2.3952}{2}}$$

$$= 1.0943$$

$$\therefore R_y = \overline{R} \pm \sigma = (13.863 \pm 1.0943)\ m$$

For green filter, $\lambda = 546\ nm$

$$R_1 = \frac{D_4^2 - D_1^2}{4(4-1)\lambda} = \frac{0.7776 \times 10^{-4}}{4 \times 3 \times 546 \times 10^{-9}} = 11.868\ m$$

$$R_2 = \frac{D_6^2 - D_3^2}{4(6-3)\lambda} = \frac{0.786 \times 10^{-4}}{4 \times 3 \times 546 \times 10^{-9}} = 11.996\ m$$

$$R_3 = \frac{D_8^2 - D_5^2}{4(8-5)\lambda} = \frac{0.906 \times 10^{-4}}{4 \times 3 \times 546 \times 10^{-9}} = 13.828\ m$$

$$\bar{R} = \frac{R_1 + R_2 + R_3}{3} = \left(\frac{11.868 + 11.996 + 13.828}{3}\right) m$$

$$= 12.564\ m$$

$$d_1 = \bar{R} - R_1 = (12.564 - 11.868) = (0.696)\ m$$

$$d_2 = \bar{R} - R_2 = (12.564 - 11.996) = (0.568)\ m$$

$$d_3 = \bar{R} - R_3 = (12.564 - 13.828) = -1.264\ m$$

$$S^2 = \frac{\Sigma d_i^2}{n} = \frac{(0.696)^2 + (0.568)^2 + (-1.264)^2}{3}$$

$$= 0.8016$$

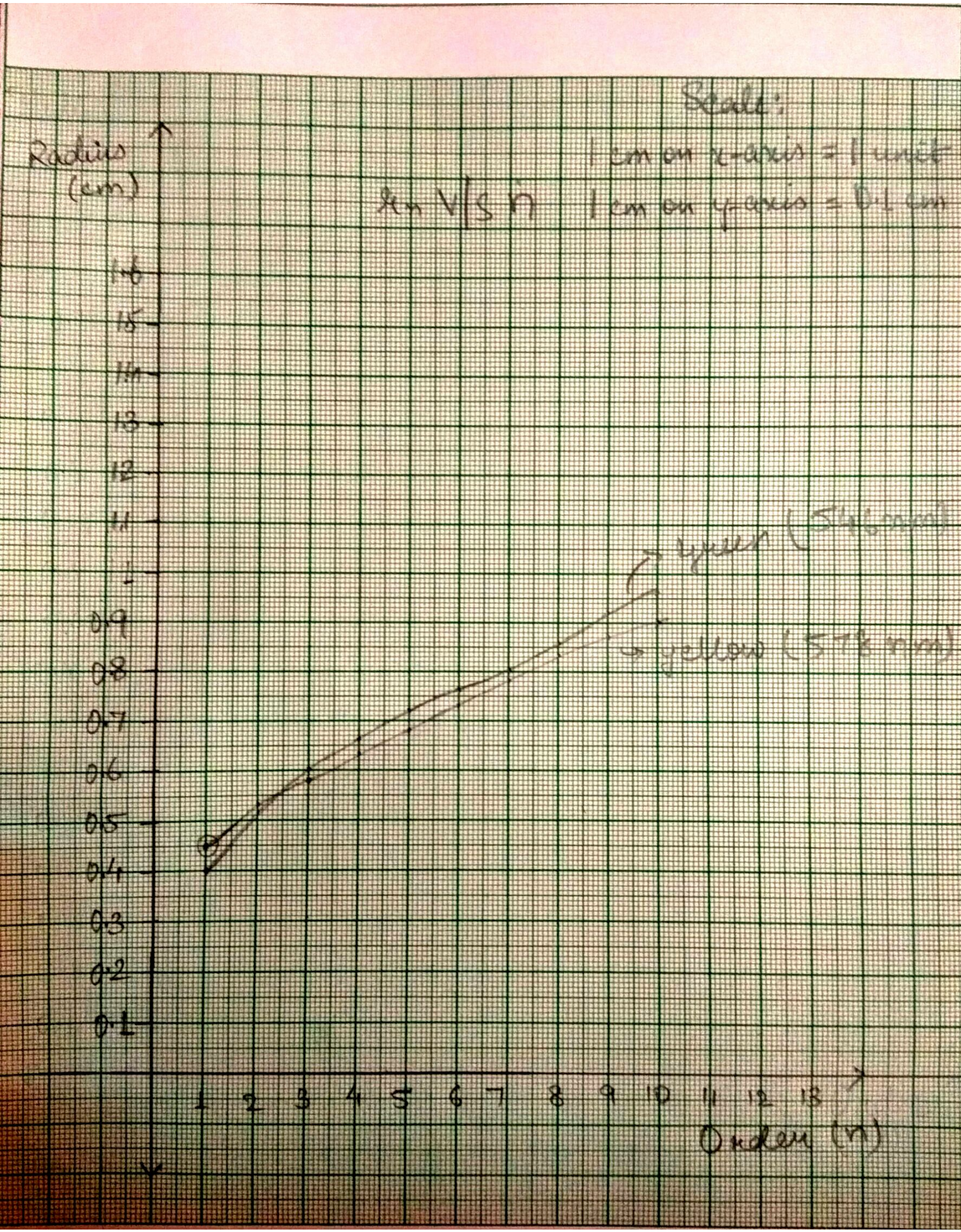
$$\sigma = \sqrt{\frac{S^2}{n-1}} = \sqrt{\frac{0.8016}{2}}$$

$$= 0.6331$$

$$\therefore R_g = \bar{R} + \sigma = (12.564 \pm 0.6331)\ m$$

Conclusion:

1. The Radius of Curvature so obtained for yellow filter is (13.863 ± 1.0943) m.
2. The Radius of Curvature so obtained for green filter is (12.564 ± 0.6331) m.
3. It can also be used to determine the wavelength of monochromatic source.
4. This experiment can be further modified to find the refractive index of liquid.
5. In this experiment air acts as the thin film.


Scale:
1cm on x-axis = 1 unit
1cm on y-axis = 0.1 cm
Radius (cm)
1.6
1.5
1.4
1.3
1.2
1.1
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
1 2 3 4 5 6 7 8 9 10 11 12 13
Order (n)
green (546 nm)
yellow (578 nm)